

Q1 - Ideal gas: It takes 80 cubic feet of air at atmospheric pressure to fill a SCUBA tank. After filling, the pressure goes up by a factor of 200. A) Roughly how much more does a full tank weigh than an empty one (please answer in kg)? B) What is the volume in litres of the interior of a tank? C) (challenge, would not be on a quiz): How much work did we do filling the tank? SCUBA tanks are always placed in water when filled so all that work has somewhere to go. Otherwise the tanks would seriously overheat.

Q2 - Units: Newton's law of gravitation says that $F = GM_1M_2/R^2$, where F is a force, G is a constant, M_1 and M_2 are masses, and R is a distance. A) What are the units of G ? please express using only kg, metres, and seconds. B) A cloud of gas has radius R and density ρ (units kg/m^3). About how long would it take to collapse? Hint - take G , the density, and the radius and try to make a time. You may or may not need all the information I have given you. C) (challenge, would never be on quiz/test): the apparent diameter of the sun is about 0.01 radians, and it takes a year for the Earth to go around the sun. What is the average density of the sun?

Q3 - sin/cos: Half angle formula. A) Express $\cos(2\theta)$ only in terms of $\cos(\theta)$ and constants (i.e. eliminate any $\sin(\theta)$ terms). Hint, $2\theta = \theta + \theta$. B) Express $\cos(\theta)$ only in terms of $\cos(2\theta)$ and constants. And don't forget that $\sin^2 + \cos^2 = 1$.

Possibly useful quantities: proton mass is $1/(6.e26)$ kg. There are about 30 litres in a cubic foot. $k_B = 1.38e - 23$ Joules/Kelvin. Room temperature is around 300K. Newton's constant G is $6.67e-11$ Nm^2/kg^2 .